be, in this chapter I'll describe how far we've actually gone, in both theory and practice, toward realizing teleporters and time machines, and what it would take to go further and attain control over space and time.

Teleportation in a Quantum World

In conventional science fiction depictions, a *teleporter* (or, in *Star Trek* lingo, a *transporter*) scans an object to determine its detailed composition and sends the information to a distant location, where the object is reconstituted. Whether the object itself is "dematerialized," its atoms and molecules being sent along with the blueprint for putting them back together, or whether atoms and molecules located at the receiving end are used to build an exact replica of the object, varies from one fictional incarnation to another. As we'll see, the scientific approach to teleportation developed over the last decade is closer in spirit to the latter category, and this raises two essential questions. The first is a standard but thorny philosophical conundrum: When, if ever, should an exact replica be identified, called, considered, or treated as if it were the original? The second is the question of whether it's possible, even in principle, to examine an object and determine its composition with complete accuracy so that we can draw up a perfect blueprint with which to reconstitute it.

In a universe governed by the laws of classical physics, the answer to the second question would be yes. In principle, the attributes of every particle making up an object—each particle's identity, position, velocity, and so on—could be measured with total precision, transmitted to a distant location, and used as an instruction manual for recreating the object. Doing this for an object composed of more than just a handful of elementary particles would be laughably beyond reach, but in a classical universe, the obstacle would be complexity, not physics.

In a universe governed by the laws of quantum physics—our universe—the situation is far more subtle. We've learned that the act of measurement coaxes one of the myriad potential attributes of an object to snap out of the quantum haze and take on a definite value. When we observe a particle, for example, the definite features we see do not generally reflect the fuzzy quantum mixture of attributes it had a moment